

FOR: Secretary Alfonso Cusi

THRU: Undersecretary Jesus Cristino Posadas

FROM: Dr. Roger D. Posadas

SUBJECT: FIRST DRAFT NUCLEAR ENERGY POLICY

DATE: 06 September 2016

Attached is the First Draft "Nuclear Energy Policy for the Republic of the Philippines"

For your review and consideration, Please feel free to call me if you have any questions about this draft policy.

Thank you and best regards.

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REPUBLIC OF THE PHILIPPINES

DEPARTMENT OF ENERGY

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(DRAFT) NUCLEAR ENERGY POLICY FOR THE

REPUBLIC OF THE PHILIPPINES

September 2016

I. POLICY PREFACE

A. Policy Purpose: This policy document presents a policy framework within which the development and utilization of nuclear energy for peaceful purposes by the Philippines will take place

B. Policy Vision: This policy envisions the creation of a modern state-of-the-art nuclear technology industry in the Philippines that will not only provide national energy security through the supply of cheap, reliable, environmentally sustainable base-load electricity needed to power our country's massive industrialization and economic development but also ensure national energy self-reliance through the technological capability to design, manufacture, construct, and commercialize globally competitive nuclear energy systems, reactors, and equipment.

C. Policy Background:

1. 1. The power situation and projection of the country. Based on existing power supply and demand projections, the Department of Energy (DOE) estimates that power shortages will affect the Visayas this year, Luzon by 2010, and Mindanao by 2011 unless new generation capacity is added. The demand for electricity is expected to increase even further in light of the country's growing population, currently pegged at 2 percent per year. Moreover, power rates in the Philippines are among the highest in Asia which can largely be attributed to our country's continuing dependence on imported fossil fuels for as much as 50% of its power generation. Besides being expensive to run, coal and oil- powered plants also release significant amounts of carbon dioxide (CO₂) and other greenhouse gases (GhGs) that have been known to cause climate change. Like many developing island nations, the Philippines is considered to be particularly vulnerable to the various negative effects associated with climate change, such as changing weather patterns and rising sea levels.

2. ***The Search for renewable energy sources***— This energy insecurity has spurred the government to search for new ways to significantly decrease the country's reliance on imported fossil fuels in order to address the looming power shortage and bring down the current price of electricity while simultaneously fulfilling global commitments to significantly reduce greenhouse gas emissions.
3. ***The Renewable Energy Act*** —in December 2008, the Philippine government enacted the Renewable Energy Act to promote the development of renewable energy resources (such as wind and solar technologies) in the country by providing significant incentives for companies to come in and invest in that sector. Yet even before the passage of the the Renewable Energy law, the Philippines was already the second largest producer of *geothermal* power in the world, and the government had already planned to continue pursuing expansion in this area. The Department of Energy (DoE) had estimated that the geothermal sector could potentially add up to 699.4 MW of capacity by 2014 but the further development of new geothermal power plants has been curtailed by the fact that most of the country's untapped geothermal steam fields are located in mountainous areas that are covered by existing ancestral domain laws.
4. ***Limitations of Geothermal and Hydroelectric Power***--. the further development of new geothermal power plants in the country has been curtailed by the fact that most of the country's untapped geothermal steam fields are located in mountainous areas that are covered by existing ancestral domain laws. Similarly, interest in the further development of large-scale hydropower projects is limited by the high initial investment cost as well as the environmental and social costs associated with the construction of large dams.
5. ***Limitations of wind power, solar power , and biomass energy***— Although the Philippines has vast *wind* energy potential, the intermittent nature of wind power, affects system operability and stability, requiring additional ancillary services and higher electricity costs. As to biomass energy, mostly from rice hull and bagasse, expanded utilization of biomass w Renewable Energy Law would be enough to encourage new investors into the sector. Finally, photovoltaic *solar* power for electricity in the Philippines is still on a very limited scale, mostly in remote, off-grid communities. The expanded use of PV solar power has been hindered by the high cost of initial investment as well as the large amount

of land required for utility-level solar power plants and possible long-term sustainability issues for these projects, particularly with regard to community maintenance and replacement of parts, which indicate that solar power in the Philippines might be appropriate only for either small-scale applications in urban areas or missionary electrification in rural areas. Solar power and wind power are also hampered by their inability to generate cheap, stable, and continuous baseload electricity given the intermittent supply of their energy resources.

6. ***The Philippine government's renewed interest in nuclear energy.*** In view of the limited ability of the country's renewable energy resources to supply cheap, stable, and continuous baseload electric power, growing concerns over the availability, high cost, and sustainability of electricity in the country have renewed the Philippine government's interest in nuclear energy, particularly the rehabilitation and re-commissioning of the mothballed Bataan Nuclear Power Plant (BNPP) in Morong, Bataan.
7. *Philippine request for an IAEA Review Mission* for an initial assessment of the BNPP-- in 2008, upon the request of the Philippine government, an 8-man mission was sent by the IAEA to the Philippines for the initial assessment and discussion of the future of the BNPP.
8. ***Findings and Recommendations of the IAEA Mission*** -- the IAEA Mission estimated that it would cost the Philippine government a minimum US\$800 million and take at least five (5) years to rehabilitate the plant's infrastructure and update its operating systems. This IAEA cost estimate of rehabilitating the BNPP is enough to fund the construction of a brand new fossil fuel-based power plant. Among the IAEA Mission's major recommendations were (1) the Drafting of a Nuclear Energy Policy, (2), the Establishment of a Dedicated Philippine Government Core Group on Nuclear Energy, and the (3) Preparation of a Feasibility Study for the Rehabilitation of the BNPP.
9. ***The need for a Nuclear Energy Policy -- This document is precisely intended to address the country's lack of a Nuclear Energy Policy*** to guide the country's development and utilization of nuclear energy, as recommended by the IAEA Review Mission in 2008,
10. ***The establishment of a Dedicated Government Core Group***—In 2009, the DoE Secretary issued Joint Department Order No. . *The establishment of a Dedicated Government Core Group*—In 2009, the DoE Secretary issued Joint Department Order No. JDO 2009-01-001 creating

the Inter-Agency Core Group on Nuclear Energy which is chaired by DoE with NPC and DOST as members and tasked to manage the development of the country's nuclear energy program and to address various infrastructure milestones and issues and to look at the feasibility of re-commissioning the BNPP.

11.. *KEPCO's Feasibility Study on the Rehabilitation and Recommissioning of BNPP*—In 2009 KEPCO conducted a Feasibility Study of the BNPP under a MOU between NPC and KEPCO. In January 2010 KEPCO submitted its Final Feasibility Report on the BNPP and stated that (a) 80% of the BNPP's equipment need overhauling and inspection and (b) the remaining 20% need replacement. In conclusion, KEPCO reported in 2010 that BNPP can be successfully rehabilitated and re-commissioned but this would cost about US\$1Billion and take four (4) years.

12. *The current legislative framework for the promotion and regulation of nuclear technology in the Philippines* is severely outdated since it is based on the Science Act of 1958 (RA 2067) and the Atomic Energy Regulatory and Liability Act of 1968 (RA 5207), which provides for two separate regulatory authorities governing the use of radiation. The PNRI, under the Department of Science and Technology (DOST) regulates of nuclear and radioactive materials used in all fields (including medical) while the Bureau of Health Devices and Technology (BHDT) under the Department of Health (DOH) regulates electrically generated radiation emitting devices used in all fields (including industry). Having two separate regulatory authorities creates space for differing regulatory policies and safety standards and also creates confusing and conflicting situations with regard to defining regulatory responsibilities and areas of implementation.

13. *The need for a modern, rational and comprehensive legislative framework support and regulate nuclear industry*

14. *Legislative Initiatives*-- two bills (Senate (Senate Bill No. 2665 authored by Sen. Miriam Defensor-Santiago and House Bill No. 4631 authored by Rep. Mark Cojuangco) are currently filed in both houses of Congress calling for the immediate rehabilitation and operation of the mothballed BNPP. Under the proposed bills, the rehabilitation and subsequent operation of the BNPP will be carried out by Napocor under the supervision and control of the DoE and the PNRI. For the first 10 years of operation, the nationality requirements for filling up the scientific/technical, supervisory, and managerial positions required by the BNPP will be

waived while the country trains and cultivates a pool of local skilled manpower for managing and operating the BNPP and other future NPPs. The amount needed for the initial implementation of the proposed bills will be charged against the appropriations of the DoE under the General Appropriations Act (GAA), or appropriated and covered by the Napocor in its annual budget. The Napocor shall also collect US\$0.1-0.2/kwh to fund the decommissioning of the BNPP when the plant eventually reaches the end of its operational life.

15. *International Obligations on Nuclear Safety Needing Senate*

ratification -- there are still a number of international conventions and treaties on nuclear safety that the Philippine Senate needs to ratify to ensure our country's commitment to the highest international standards of nuclear safety and security. . Ratification of these conventions will commit the country towards implementing high levels of nuclear safety and security. States that have ratified these conventions will most likely refuse to provide nuclear materials, fuel, equipment and expertise to the Philippines if it does not ratify and comply with said conventions.

D. Policy International Context : Nuclear energy is becoming an increasingly attractive option for the future given its ability to provide stable and continuous baseload power while significantly avoiding the release of harmful GhGs into the atmosphere. Nuclear energy is currently being used in most developed countries, with a number of developing countries also considering it. As of July 2008, there were 430 Nuclear Power Plants or NPPs operating worldwide. Electricity generated by these plants accounts for 16 percent of total global supply. In the US, 104 NPPs provide almost 20 percent of the country's electricity. Almost a third of Western Europe's power supply is generated by around 150 NPPs. In France alone, 59 NPPs generate 80 percent of the country's power. Among the G8 (the world's eight largest economies), Italy is the only country without its own nuclear power program. It is also the world's largest net importer of electricity. As such, the Italian government confirmed last year that it will begin building new nuclear plants within the next five years in order to reduce the country's dependence on imported power. Many of the Philippines' neighbors are also considering the nuclear option. Vietnam passed its own Atomic Energy Act in 2008. Construction of their first four-turbine NPP will begin in 2015 and will be completed in 2025. Indonesia also has plans of having three NPPs up and running within the next two decades, with the first one in operation by 2015. Thailand is also conducting studies to assess the compatibility of nuclear

E. Policy Objectives:

1. **Promotion of Nuclear Energy as a means of supplying cheap, reliable, environmentally clean baseload electricity**
2. **Creation of a modern and comprehensive legislative and regulatory framework for the safe and secure utilization of nuclear energy**
3. **Establishment of a nuclear energy governance framework to clarify the roles and responsibilities of various players in a nuclear energy industry.**
4. **Promotion of energy security, environmental sustainability and scientific and technological self-reliance in nuclear energy .**

II. NUCLEAR ENERGY POLICY FRAMEWORK

A. Policy Principles:

1. **All Nuclear power operations shall be subjected to Complete Operational Transparency.**
2. **All nuclear energy-related activities shall be subjected to the highest international standards of safety and security**
3. **Nuclear energy shall be used only for peaceful purposes and in compliance with international legal obligations and commitments.**
4. **The development and utilization of energy shall be undertaken in cooperation with the IAEA and within a legal regulatory framework that is consistent with international standards and best practice and in conformity with international legal obligations and commitments.**
5. **The acquisition and development of nuclear power capability shall be carried out in partnership with governments and firms of responsible nations and with the technical assistance of appropriate international nuclear energy experts**
6. **The development and Implementation of a nuclear power program shall be pursued in a way that best ensures national energy security and environmental sustainability**
7. **The acquisition and development of nuclear technology shall be undertaken on the basis of technonationalist principles that seek to achieve a self-reliant technological capability in the design, development, construction, and commercialization of Philippine nuclear reactors, equipment, fuels, and related systems.**

8. A national program for the world-class education and training of nuclear scientists, engineers, and technicians shall be implemented by the government to provide a large pool of nuclear energy professionals and technologists needed to meet the human resource requirements of a modern nuclear industry in the Philippines.
9. A national program of research, development, and innovation shall also be undertaken by the Philippine government in coordination with Philippine universities and firms to buttress the development, utilization and deployment of nuclear energy technology in the Philippines and to acquire technological know-how and capability in next-generation nuclear reactors, thorium-fuelled nuclear reactors and even nuclear fusion reactors. r scientists, engineers, and technicians
10. A national program to inform the public about nuclear energy and in particular, about the BNPP shall be implemented by the DoE through various media.
11. Nuclear Power Plants (NPPs) in the Philippines shall be operated by a government- owned and controlled corporation although under special circumstances, an NPP may be operated under a public-Private Partnership or Joint Ventiure provided controlling shares are retained by the government-owned and controlled nuclear power corporation

III. NUCLEAR ENERGY GOVERNANCE FRAMEWORK

- A. Institutional Arrangements: The following orgaizations/institutions shall be established to ensure the effective implementation of this nuclear energy policy:
 1. An Inter-governmental organization for National Executive Coordination which shall ensure implementation and exercise oversight over all aspects of this nuclear energy policy.
 2. A Presidential Commission on the Bataan Nuclear Power Plant (PCBNPP) which shall be the coordinating , policy-making and implementing body on all BNPP issues, including management, technical and financial policies/activities related to the rehabilitation and re-commissioning of the BNPP.
 3. An independent organization for nuclear energy regulation to be called Philippine Nuclear Regulatory Authority (PNRA)

4. A government-owned and controlled corporation to manage the operation of the BNPP and to be called Philippine Nuclear Power Corporation (PNPC).
5. An organization for radioactive waste management to be called Philippine Radioactive Waste Management Agency (PRWA).
6. A national organization for nuclear energy research development, and innovation to be called National Institute for Nuclear Energy Research, Development and Innovation (NINERDI)
7. A national organization for nuclear reactor architecture, civil engineering, construction, and equipment manufacturing to be called Philippine Nuclear Architecture and Construction Company (PNACC)..

IV. NUCLEAR ENERGY LEGISLATIVE FRAMEWORK

- V. A. Review and Consolidation of current Senate and House Bills on Nuclear Energy and on the rehabilitation and operation of the BNPP.
- B. Enactment of updated laws and executive for the following:
 1. The creation / establishment of an independent and effective national regulatory authority for the development and utilization of nuclear energy to be called Philippine Nuclear Regulatory Authority (PNRA)
 2. The creation/ establishment of a national radioactive waste management agency to be called Philippine Radioactive Waste Management Agency (PRWMA)
 3. the establishment of a nuclear liability regime for third party compensation.
 4. the establishment of a government-owned and controlled corporation for electricity generation from nuclear power to be called Philippine Nuclear Power Corporation (PNPC).

5. The establishment of a national institute for nuclear energy research, development, and innovation to be called National Institute for Nuclear Energy Research, Development, and Innovation (NINERDI)
 6. The establishment of a government-owned and controlled corporation for nuclear reactor architecture, civil engineering and Fabrication to be called Philippine Nuclear Architecture and Construction Company. (PNACC).
 7. The establishment of an integrated national system of educational and training programs and associated scholarships for the development of a large national pool of world-class nuclear scientists, engineers, technologists, technicians, and technology managers.
 8. The regulation of Public-Private Partnerships and technology transfer in the Philippine nuclear energy industry.
- C. Senate Ratification of the following International conventions and treaties on nuclear safety to commit the Philippines towards implementing high levels of nuclear safety and security because States that have ratified these conventions will most likely refuse to provide nuclear materials, fuel, equipment and expertise to the Philippines if it does not ratify and comply with said conventions. These international conventions include the following;
1. *Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management which* applies to spent fuel and radioactive waste resulting from civilian nuclear reactors and applications; and to spent fuel and radioactive waste from military or defense programs if and when such materials are transferred permanently to and managed within exclusively civilian programs, or when declared as spent fuel or radioactive waste for the purpose of the Convention by the Contracting Party. The Convention also applies to planned and controlled releases into the environment of liquid or gaseous radioactive materials from regulated nuclear facilities.
 2. *Convention on Nuclear Safety which* aims to legally commit participating States operating land-based nuclear power plants to maintain a high level of safety by setting international benchmarks to which these States would subscribe
 3. *Convention on Supplementary Compensation for Nuclear Damage*
 - - This Convention requires compensation for nuclear damage for which an operator of a nuclear installation situated in the territory

of a Contracting Party is liable under either international conventions or national law.

4. *Joint Protocol Relating to the Application of the Vienna Convention and the Paris Convention* - - The Joint Protocol combines the Vienna and Paris Conventions, both of which concern the liabilities of parties with regard to nuclear damage. Parties to the Joint Protocol are treated as though they are Parties to both Conventions and a choice of law rule is provided to determine which of the two conventions should apply to the exclusion of the other in respect of the same incident.
5. Additional Protocol to the Safeguards Agreement- This would grant the IAEA complementary inspection authority to those provided in underlying safeguards agreements. The principal aim is to enable the IAEA inspectorate to provide assurances about both declared and possibly undeclared activities. Under the Protocol, the IAEA is granted expanded rights of access to information and sites.

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V. NUCLEAR ENERGY POLICY IMPLEMENTATION TIMELINE

A. Phase 1 (Sept 2016-Sept 2017):

1. Approval of the Nuclear Energy Policy.
2. Enactment by Congress of the Nuclear Energy Legislative Framework
3. Invitation by the PCBNPP/DoE of an Inspection team from GEN Energija, the joint owner/operator of Krsko, (the NPP in Slovenia, which is identical to BNPP) to conduct a detailed inspection of the technical and safety conditions of BNPP's operational and supporting equipment and facilities as well as its safety compliance and human resource needs.
4. Dispatch of a DoE/PCBNPP observation team to the KRSKO in Slovenia and Angra in Brazil to observe these successfully operating Westinghouse-designed pressurized water reactor NPPs which are identical to the BNPP.
5. Establishment of the Philippine Nuclear Regulatory Authority (PNRA)
6. Request by PNRA to IAEA to dispatch an Operational Safety Review Team (OSART) to conduct an in-depth review of the operational safety performance and operational readiness of BNPP.

B. Phase 2 (Sept 2017-Sept 2018:)

- 1. Conduct by GEN of a detailed inspection of the BNPP**
- 2. Establishment of the Philippine Nuclear Power Corporation (PNPC) to manage the operation of the BNPP.**
- 3. Submission by GEN of its Inspection Report on BNPP with a detailed start-up plan and investment plan for de-conservation, refurbishing, technical upgrading, and lifespan/capacity improvements.**
- 4. Go.No Go Decision to Rehabilitate and re-commission BNPP.**
- 5. Start of Rehabilitation of the BNPP.**
- 6. Rehiring of Trained Filipino nuclear scientists, engineers, and technicians who have gone abroad or into retirement after the mothballing of the BNPP but who are willing to come back and work in BNPP if it is re-commissioned.**
- 7. Sending of Filipino scientists, engineers, and technicians to KRSKO in Slovenia for training in the regulation, operation and management of an NPP identical to the BNPP.**
- 8. Sending of personnel of the Philippine Nuclear Regulatory Authority for training by experienced nuclear regulatory agencies abroad**

C. Phase 3 (Sept 2018-Sept 2019):

- 1. Completion of BNPP's Rehabilitation.**
- 2. Formation and registration of PhilGEN as a Joint Venture between GEN and the Philippine NuclearPower Corporation (PNPC) for the re-start and operation of BNPP.**
- 3. Conduct by the IAEA OSART of an in-depth review of the operational readiness of BNPP.**
- 4. Submission of OSART Report with recommendations.**
- 5. Hot functional Tests of BNPP and initial synchronization to the grid**
- 6. Delivery of Nuclear Fuel to BNPP.**
- 7. Conduct by PNRA of public hearings on the licensing of the BNPP.**

8. Licensing of PhilGEN to restart and operate the BNPP.

9. Refuelling of BNPP.

10. Re-start of the BNPP.

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